What explains movement across the AI value chain?

Can a compact, interpretable factor set explain AI-linked stocks - and what fit is lost when we simplify?

COMPACT MODEL EVALUATED

60 targets

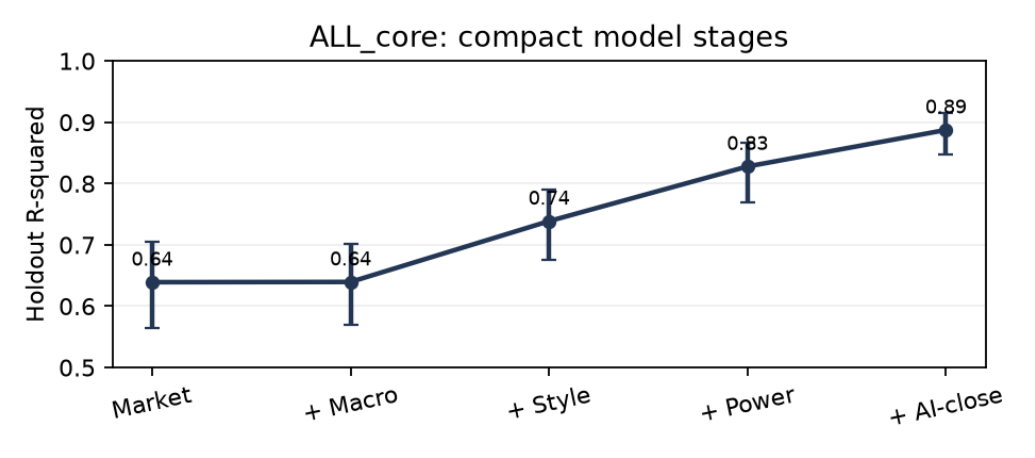
47 stocks + 13 baskets

DIRECT LIKE-FOR-LIKE TEST

58 targets

47 stocks + 11 common baskets

COMPACT MODEL: FIT ADDED IN STAGES



A transparent comparison of the original 12-factor model and a five-factor baseline. R-squared measures fit, not forecast direction or causality.

The data is downloaded, cleaned and checked

Daily price history feeds one reusable pipeline; suspicious observations are flagged for review, not silently deleted.

TICKERS DOWNLOADED

621 / 623

2 returned empty

PRICE HISTORY

2016-2026

Through 1 Oct 2026

ISSUES FLAGGED

~5,100

Outliers and stale prices

SANITY CHECKS

6 / 6 pass

Funds vs. benchmarks

FUND / INDEX CORRELATION CHECKS

Observed daily-return correlation; closer to 1 means closer tracking.



What this supports

The benchmark checks suggest the adjusted price series behave consistently with their tracked indexes. Around 5,100 unusual or stale observations were marked for transparent review.

Download -> clean and adjust -> quality checks -> factor and model analysis

Seven layers connect the AI value chain

The stock basket follows the infrastructure behind AI, from compute to electricity supply.

AI-LINKED STOCK LAYERS

Chips	NVDA, AMD, AVGO, TSM, MU
Networking / servers	ANET, SMCI, DELL, CLS
Hyperscalers	MSFT, GOOGL, AMZN, META
Data centers	EQIX, DLR, CRWV, NBIS
Power generation	VST, CEG, NRG, TLN
Grid / electrical	ETN, VRT, PWR, GEV
Nuclear / uranium	CCJ, BWXT, LEU, OKLO

39 AI-chain stocks in the original basket

>

12 FACTORS IN THREE INTERPRETABLE GROUPS

MARKET / MACRO / STYLE

MKT | D10Y | USD | OIL | LOWVOL

AI-ADJACENT INFRASTRUCTURE

POWER | GRID | URANIUM

AI-CLOSE

SEMI | SOFT | ROBO | MOM

The compact model keeps one representative per factor type.

Redundant candidates were grouped before modeling

Correlation clustering and variance-inflation checks narrowed the library; Gram-Schmidt orthogonalization then separated retained signals.

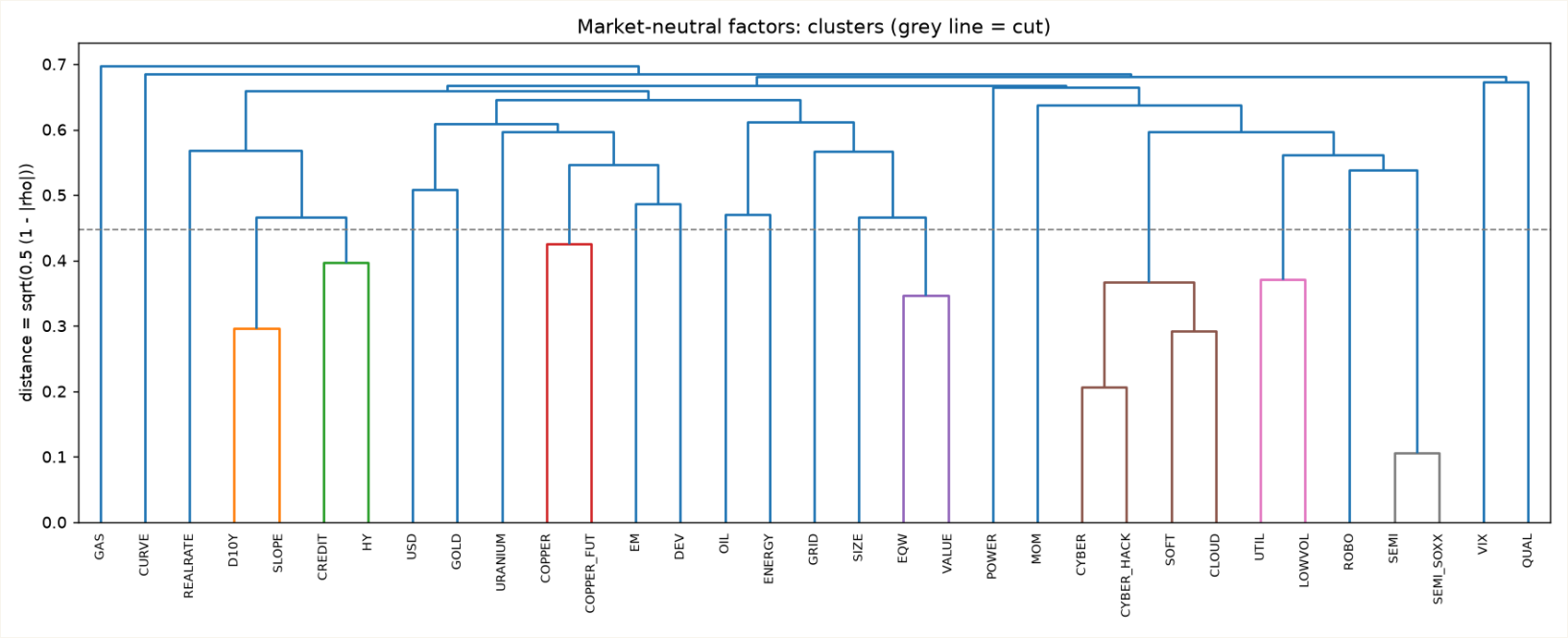
CANDIDATE FACTORS
38
Screened for the model

RETAINED FACTORS
12
After clustering / VIF

CONDITION NUMBER
354 -> ~20
Lower collinearity

MAXIMUM VIF
4.5
After selection

SIMILAR FACTOR RETURNS FORM CLUSTERS; THE DASHED LINE MARKS THE CUT USED HERE



The cluster tree illustrates similarity, not economic causality. VIF and condition-number checks help limit unstable overlap.

Where did the 12-factor model gain fit?

Out-of-sample R-squared by target layer as factor blocks are added (2024-2026 holdout).

	Market only	+ Macro / style	+ AI-adjacent	+ AI-close
Chips	0.57	0.70	0.76	0.91
Networking / servers	0.45	0.55	0.65	0.71
Hyperscalers	0.61	0.65	0.67	0.74
Data centers	0.27	0.31	0.40	0.40
Power generation	0.26	0.30	0.83	0.84
Grid / electrical	0.41	0.47	0.76	0.77
Nuclear / uranium	0.24	0.27	0.68	0.68
Core basket	0.64	0.74	0.89	0.92

Infrastructure layers gained most from AI-adjacent factors; chips gained most from AI-close factors. Data centers remained difficult to explain (R-squared 0.40).

Compact-model fit across the same seven layers

Out-of-sample R-squared by factor stage, using the same target layers as the original-model table.

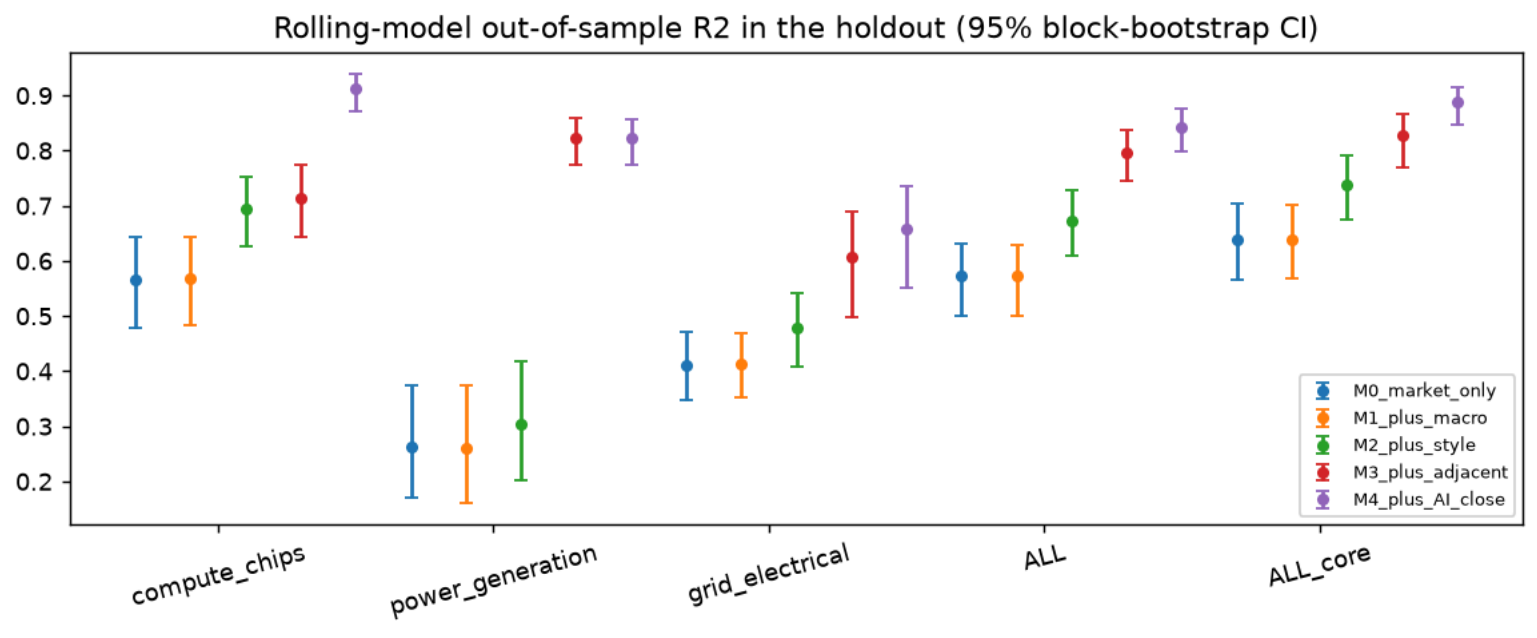
TARGET LAYER	M0 Market only	M1 + D10Y	M2 + LOWVOL	M3 + POWER	M4 + SEMI
Chips	0.57	0.57	0.69	0.71	0.91
Networking / servers	0.45	0.45	0.55	0.61	0.70
Hyperscalers	0.61	0.61	0.66	0.65	0.70
Data centers	0.26	0.26	0.31	0.35	0.36
Power generation	0.26	0.26	0.30	0.82	0.82
Grid / electrical	0.41	0.41	0.48	0.61	0.66
Nuclear / uranium	0.24	0.24	0.27	0.36	0.37
Core basket	0.64	0.64	0.74	0.83	0.89

Values are point estimates; R-squared measures fit, not directional accuracy or paired significance.

CORE BASKET	CHIPS	POWER
0.64 -> 0.89	+0.20 at SEMI	+0.52 at POWER

Five factors, added in interpretable steps

Same rolling-regression framework and holdout dates as the larger model.



THE FACTOR LADDER

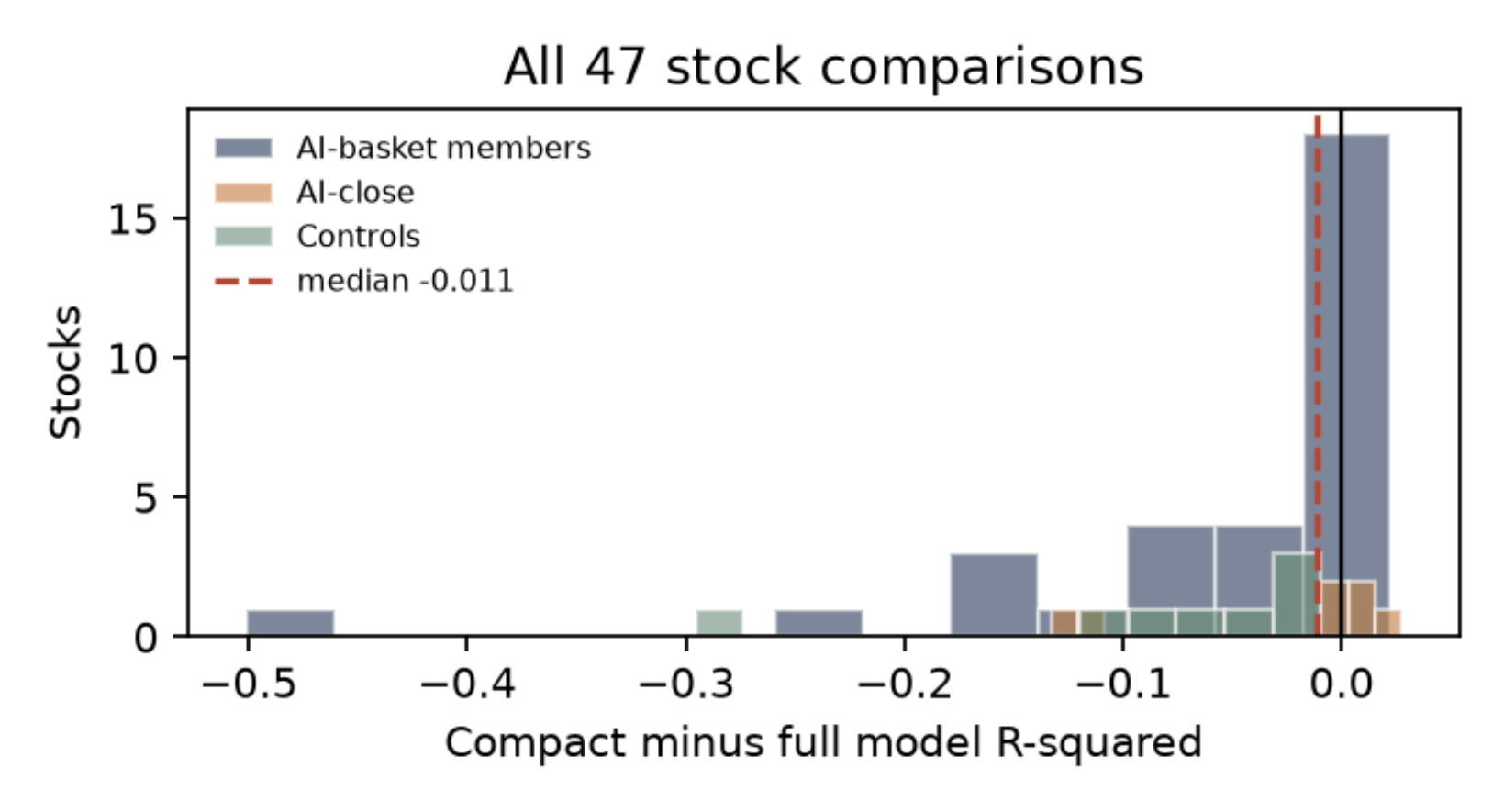
0	M0	MKT
1	M1	+ D10Y
2	M2	+ LOWVOL
3	M3	+ POWER
4	M4	+ SEMI

Broad-basket R-squared: 0.64 -> 0.89. D10Y adds little alone; later stages add more.

Evaluation: estimate on 2017-2023; test on 2024-2026; rolling window = 252 trading days.

Stock-by-stock comparison: compact vs. full model

How did the five-factor model compare with the 12-factor model across individual stocks?



Negative differences favor the full model; positive differences favor the compact model. Median difference: -0.011 R-squared.

The compact model is close for many stocks, but not all

Each plotted point compares the two models on the same target and holdout period.

STOCKS COMPARED

47 / 47

Same stock definitions

COMPACT SCORE HIGHER

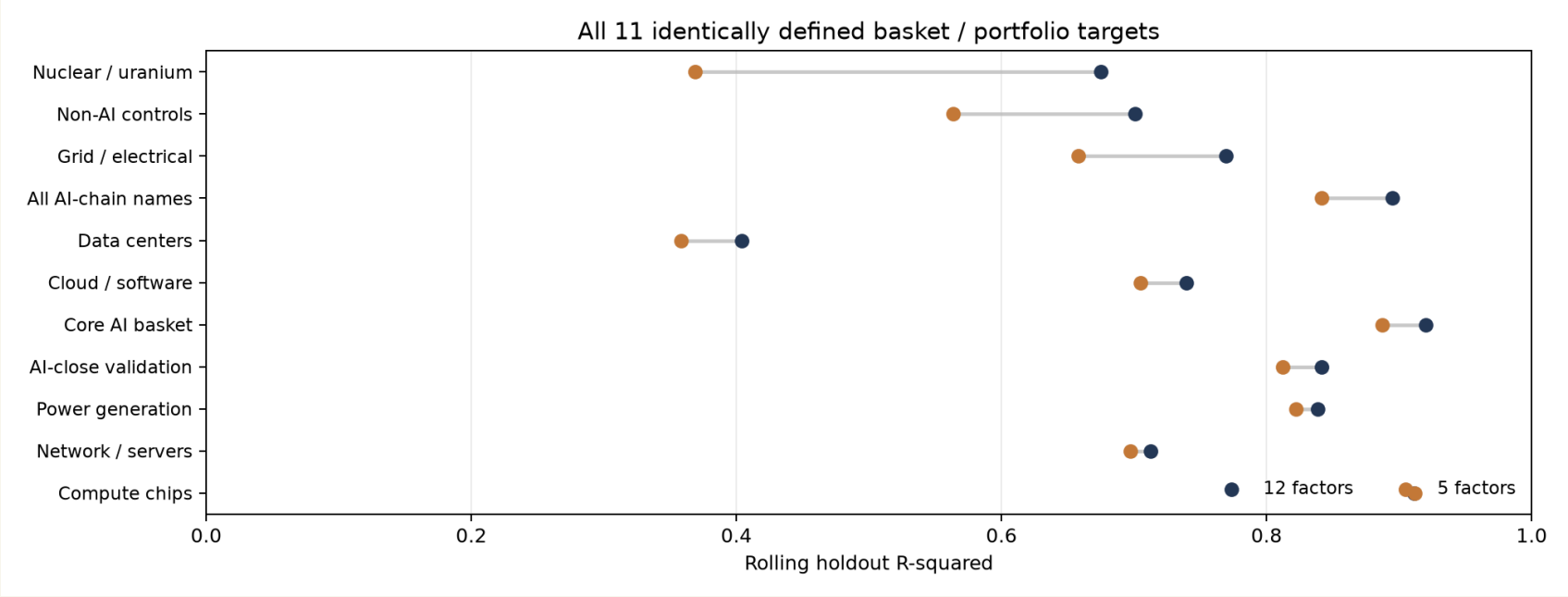
15 / 47

Median difference: -0.011 R-squared

CLOSE TO FULL MODEL

31 / 47

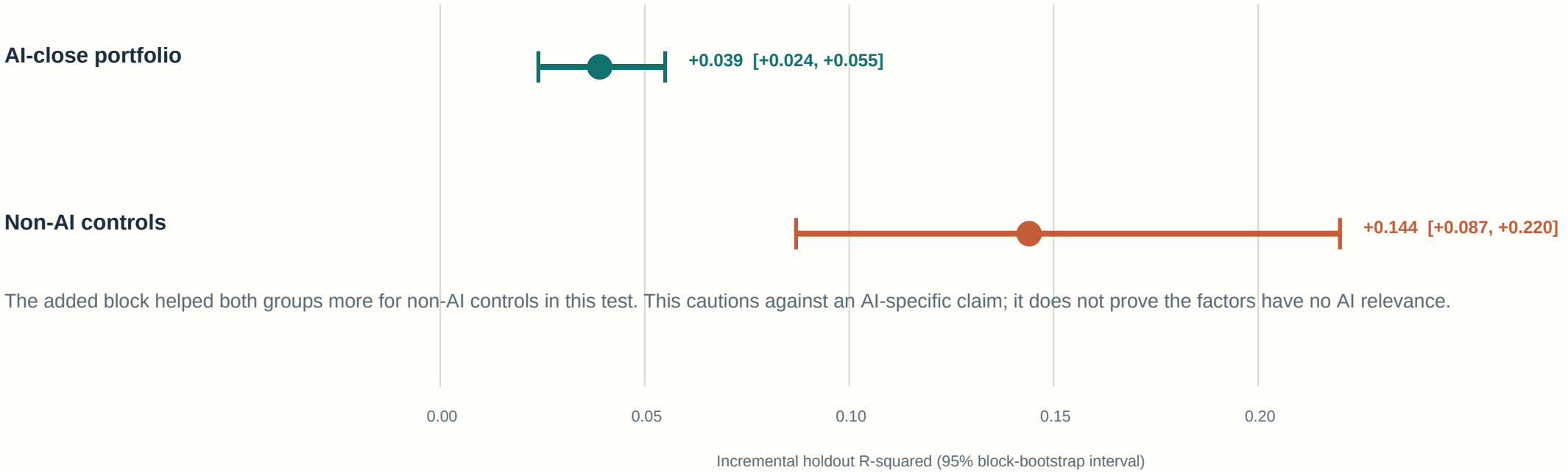
Within 0.05 R-squared



Compact model scored 60 targets; like-for-like comparison covers 58 (47 stocks + 11 common baskets). Overlap-excluded baskets have different constituents and are excluded.

The AI-close block is not AI-specific yet

Incremental R-squared when adding AI-close factors to the adjacent-factor model (M3 - M2).

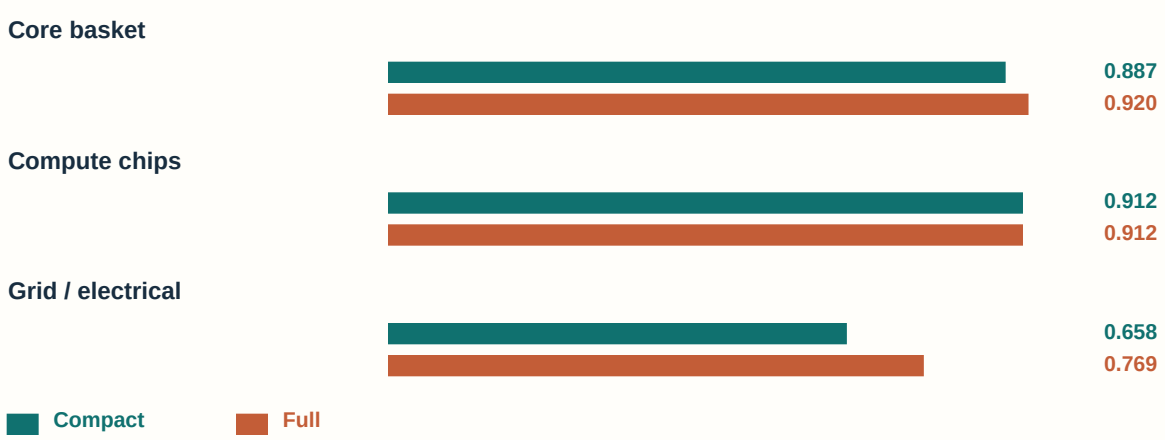


The added block helped both groups more for non-AI controls in this test. This cautions against an AI-specific claim; it does not prove the factors have no AI relevance.

Use the compact model as a baseline, not a replacement

The original model retains useful detail for selected layers; evidence is informative, not causal.

HOLDOUT R-SQUARED: COMPACT VS. FULL MODEL



What the comparison says

The five-factor model is simpler and stays close on many stocks. The full model remains stronger for some infrastructure groups, especially grid / electrical.

Limits: one holdout period; some ETF factors hold target stocks; R-squared alone makes no causal or investment claim.

NEXT TESTS

1. **Exclude direct ETF holdings**
2. **Try alternative factor representatives**
3. **Repeat across additional periods**